

115. The standard deviation of 100 observations is 10. If 20 is added to each observation, then what will be the new standard deviation?

- (a) 10
- (b) 15
- (c) 20
- (d) 25

116. Let X be a random variable following binomial distribution with parameters $n = 5$ and $p = k$. Further, $P(X = 1) = 0.4096$ and $P(X = 2) = 0.2048$. What is the value of k ?

- (a) 0.2
- (b) 0.25
- (c) 0.3
- (d) 0.35

117. The frequency distribution of the marks obtained by students in a Science examination is given below :

Marks	5-15	15-25	25-35	35-45
Number of students	20	30	30	20

What is the arithmetic mean?

- (a) 20
- (b) 25
- (c) 30
- (d) 35

118. If $P(A) = 0.3$, $P(B) = 0.4$ and $P(A|B) = 0.5$, then what is the value of $P(B|A)$?

- (a) 0.325
- (b) 0.333
- (c) 0.375
- (d) 0.667

119. If $P(A) = 1/3$, $P(B) = 1/2$ and $P(A \cap B) = 1/4$, then what is the value of $P(\bar{A} \cup B)$?

- (a) $7/12$
- (b) $2/3$
- (c) $3/4$
- (d) $11/12$

120. Consider the following statements :

- I. Mean and variance have the same unit of measurement.
- II. Mean deviation and standard deviation have the same unit of measurement.

Which of the statements given above is/are correct?

- (a) I only
- (b) II only
- (c) Both I and II
- (d) Neither I nor II

SPACE FOR ROUGH WORK